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Roll No

EX-6001-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Communication Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Find the Fourier Transform of the following functions and draw the spectrum:
 - i) $g(t) = e^{-bt} u(t)$
 - ii) Constant function A
- b) Give classification of signals. How can you differentiate energy and power signals?
2. a) List of Application of central limit theorem.
- b) Give a comparison between AM, DSB-SC, SSB-SC and VSB-SC.
3. a) Draw the block diagram of typical communication system and explain each block.
- b) Draw and explain the Pre-emphasis and De-emphasis circuit with a neat diagram.
4. a) Illustrate the relationship between FM and PM with the help of block diagram.
- b) Derive an expression for an FM signal when a carrier $A \sin \omega_c t$ is being modulated by a signal $E_m \cos \omega_m t$.

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5. a) Draw block diagram of superheterodyne receiver and explain each block.
b) What is TRF receiver? Compare this with superheterodyne receiver.
6. a) What is Quantization? Explain quantization error and Non-linear quantizer.
b) Draw block diagram of PCM system and explain it comment on its bandwidth.
7. a) Draw the block diagram of a satellite system and explain its working.
b) What is the working of transponder and earth station explain in detail.
8. Write short notes on (any three):
 - i) PCM
 - ii) Transponders
 - iii) TRF receivers
 - iv) QAM
